Unit 1

Introduction to Computer Science

A Complete Course Guide

Foundations of Computers, Hardware, Software, and Digital Literacy

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LESSON 1.0

Welcome to Computer Science Class

Welcome to your first course in Computer Science! This class is designed to introduce you to the world of computers — how they work, what they are made of, and how to use them confidently and effectively. Whether you have never touched a computer before or you already use one every day, this unit will build a strong foundation for everything else you will learn in this course.

What You Will Learn

- What computer science is and why it matters in the modern world.
- The different modern types of computers and where they are used.
- The parts of a computer (hardware) and what each part does.
- The kinds of programs (software) that make computers useful.
- How to use the keyboard and mouse correctly and confidently.
- How to navigate a computer's desktop, icons, and folders.

Course Goals

By the end of this unit, you should be able to identify and explain the basic building blocks of computing, operate a computer independently for everyday tasks, and speak confidently using proper computer terminology. These foundational skills will support every future unit in this course, including programming, data, and problem solving.

Class Expectations

1. Come to class ready to participate in hands-on practice — computer science is learned by doing.
2. Complete each assignment (1.1–1.6) before attempting the final assessment.
3. Ask questions! There are no "silly" questions when you are learning something new.
4. Practice typing and mouse skills a little every day — small daily practice builds big results.
5. Be patient with yourself. Everyone learns computer skills at their own pace.

Why Study Computer Science?

Computers are part of almost every job, every industry, and everyday life — from communication and shopping to healthcare and transportation. Learning how computers work and how to use them well is one of the most valuable skills you can build today.

LESSON 1.1

Introduction to Computer Science — Modern Types of Computers

What Is a Computer?

A **computer** is an electronic device that accepts data (input), processes that data according to a set of instructions, stores it, and produces a result (output). Computers can perform millions of calculations every second, follow instructions exactly, and repeat tasks without getting tired. The basic cycle every computer follows is often summarized as:

> **The Information Processing Cycle**
>
> **Input → Process → Output → Storage**
> **Input:** Data goes into the computer (e.g., typing on a keyboard).
> **Process:** The computer's processor works on the data following instructions.
> **Output:** The result is displayed or delivered to the user (e.g., on a screen or printer).
> **Storage:** Data and results are saved for later use.

What Is Computer Science?

Computer Science is the study of computers and computational systems — how they are designed, how they process information, and how we can use them to solve problems. It includes topics such as hardware, software, programming, algorithms, data, networks, and artificial intelligence. Computer scientists design the systems and applications that power modern life, from mobile apps and websites to medical devices and spacecraft.

Modern Types of Computers

Computers come in many shapes and sizes, each designed for a different purpose. Here are the main types of computers used today:

Type	Description	Common Uses
Desktop Computer	A personal computer designed to stay in one place, with a separate monitor, keyboard, and mouse.	Offices, schools, home, gaming
Laptop Computer	A portable computer with the screen, keyboard, and components built into one foldable case.	Students, remote work, travel
Tablet	A slim, portable, touchscreen computer without a physical keyboard.	Reading, browsing, note-taking, drawing
Smartphone	A small handheld computer that also functions as a mobile phone.	Communication, apps, photography
Server	A powerful computer that provides data, services, or resources to other computers over a network.	Websites, email, cloud storage
Supercomputer	An extremely powerful computer capable of extremely fast, large-scale calculations.	Weather forecasting, scientific research
Embedded Computer	A small computer built into another device to control a specific function.	Microwaves, cars, ATMs, smart watches

Quick Check

- Can you name three types of modern computers and one use for each?
- What are the four steps in the Information Processing Cycle?
- Why do you think computer science is considered useful in almost every career field today?

LESSON 1.2

Computer Hardware — Parts of a Computer and Functions

What Is Hardware?

Hardware refers to the physical parts of a computer — anything you can see and touch. Hardware works together with software (the instructions) to make a computer function. Hardware is generally grouped into four categories: input devices, output devices, processing/storage devices, and internal components.

Core Internal Components

Component	Function
CPU (Central Processing Unit)	The "brain" of the computer; carries out instructions and performs calculations.
Motherboard	The main circuit board that connects all hardware components together.
RAM (Random Access Memory)	Temporary, high-speed memory used to store data the CPU is actively using.
Hard Drive / SSD	Permanent storage that keeps files, programs, and the operating system even when the p
Power Supply Unit (PSU)	Converts electricity from the wall outlet into usable power for the computer.
Graphics Card (GPU)	Processes and renders images, video, and animations for the display.
Cooling Fan / Heat Sink	Keeps internal components from overheating during use.

Input Devices

Input devices allow the user to send data or commands into the computer.

- **Keyboard** — used to type letters, numbers, and commands.
- **Mouse** — used to point, click, and select items on the screen.
- **Microphone** — captures sound and voice input.
- **Scanner** — converts paper documents or images into digital files.
- **Webcam** — captures video input for calls or recording.

Output Devices

Output devices display or deliver the results processed by the computer.

- **Monitor** — displays visual output such as text, images, and video.
- **Printer** — produces a paper copy of digital documents.
- **Speakers** — produce audio output such as music or notifications.
- **Headphones** — deliver private audio output to one user.

Storage Devices

- **Internal storage** (hard drive or SSD) — stores the operating system and files inside the computer.

- **External storage** (USB flash drive, external hard drive) — portable storage that can be connected and removed.
- **Cloud storage** — data saved on remote servers and accessed over the internet (e.g., Google Drive, OneDrive).

> **Remember**
>
> A helpful way to remember hardware categories: **Input** goes IN, **Output** comes OUT, and **Storage** keeps data safe for later — while the CPU and motherboard do the thinking and connecting in between.

LESSON 1.3

Computer Software — Types of Software and Their Uses

What Is Software?

Software refers to the sets of instructions, called programs, that tell computer hardware what to do. Unlike hardware, software cannot be physically touched — it exists as code stored on a computer's storage device. Software is generally divided into two major categories: system software and application software.

1. System Software

System software manages and controls the computer's hardware so that application software can run. It works in the background and is essential for the computer to operate at all.

- **Operating System (OS)** — manages hardware, memory, and files, and provides the interface users interact with. Examples: Windows, macOS, Linux, Android, iOS.
- **Device Drivers** — small programs that allow the OS to communicate with hardware such as printers or graphics cards.
- **Utility Programs** — tools that help maintain and optimize the computer, such as antivirus software, disk cleanup tools, and file compression tools.

2. Application Software

Application software (or "apps") helps users perform specific tasks. Unlike system software, application software is chosen and installed based on what the user wants to do.

Category	Examples	Typical Use
Word Processing	Microsoft Word, Google Docs	Writing letters, essays, reports
Spreadsheet	Microsoft Excel, Google Sheets	Calculations, budgets, data analysis
Presentation	Microsoft PowerPoint, Google Slides	Creating slideshows for presentations
Web Browser	Chrome, Firefox, Safari, Edge	Browsing the internet
Communication	Gmail, WhatsApp, Zoom	Email, messaging, video calls
Multimedia	VLC Media Player, Photoshop	Playing or editing audio, video, images
Educational / Games	Duolingo, Minecraft: Education Edition	Learning and entertainment

System Software vs. Application Software

System Software	Application Software
Runs the computer itself	Helps the user complete specific tasks
Starts automatically when the computer turns on	Must be opened by the user
Example: Windows 11 operating system	Example: Microsoft Word

Tip

Every computer needs an operating system before any application software can run — think of the operating system as the foundation of a house, and application software as the rooms built on top of it.

LESSON 1.4

The Computer Keyboard — Use, Keys, Functions, Typing Practice

What Is a Keyboard?

The keyboard is the most common input device used to type text, numbers, and commands into a computer. A standard keyboard is divided into several key groups, each with a different purpose.

Keyboard Key Groups

Key Group	Keys Included	Function
Alphanumeric Keys	Letters (A–Z), Numbers (0–9)	Typing words, sentences, and numbers
Function Keys	F1–F12	Perform special actions (e.g., F1 = Help, F5 = Refresh)
Navigation Keys	Arrow keys, Home, End, Page Up, Page Down	Move the cursor around the screen or document
Numeric Keypad	0–9, +, -, *, /	Quick entry of numbers, usually on the right side of the ke
Modifier Keys	Shift, Ctrl, Alt, Windows/Command key	Used in combination with other keys to perform shortcuts
Special Keys	Enter, Backspace, Delete, Tab, Caps Lock, Esc, Spacebar	Control text entry and editing

Common Key Functions

- **Enter/Return** — moves to a new line or confirms a command.
- **Backspace** — deletes the character to the left of the cursor.
- **Delete** — deletes the character to the right of the cursor.
- **Shift** — types a capital letter or the upper symbol on a key when held.
- **Caps Lock** — locks all letters to type in capitals until turned off.
- **Tab** — moves the cursor forward to the next stop or field.
- **Spacebar** — inserts a blank space between characters or words.
- **Ctrl / Alt** — used with other keys for shortcuts (e.g., Ctrl + C to copy, Ctrl + V to paste).
- **Esc (Escape)** — cancels or exits the current action or window.

Useful Keyboard Shortcuts

Shortcut	Action
Ctrl + C	Copy selected text/item
Ctrl + V	Paste copied text/item
Ctrl + X	Cut selected text/item

Shortcut	Action
Ctrl + Z	Undo the last action
Ctrl + S	Save the current file
Ctrl + P	Print the current document
Ctrl + A	Select all content

Proper Typing Posture

- Sit up straight with your feet flat on the floor.
- Keep your wrists straight and slightly raised — not resting heavily on the desk.
- Place your fingers on the "home row": left hand on A-S-D-F, right hand on J-K-L-;
- Look at the screen, not the keyboard, as you build muscle memory.

Typing Practice Exercises

Practice the following exercises to build accuracy and speed. Start slowly and focus on correct finger placement before increasing speed.

1. **Home Row Practice:** Type the following line 5 times: *asdf jkl; asdf jkl;*
2. **Word Practice:** Type each word 3 times: *sad, lad, fall, jak, all, dash, flask*
3. **Sentence Practice:** Type the sentence 5 times, focusing on accuracy: *"The quick brown fox jumps over the lazy dog."*
4. **Number Row Practice:** Type the following 5 times: *1234567890 0987654321*
5. **Shortcut Practice:** Open a blank document and practice Ctrl + C, Ctrl + V, and Ctrl + Z on a sample paragraph of text.

Practice Goal

Beginners should aim for accuracy first, then speed. A good early goal is 15–20 words per minute with at least 90% accuracy before increasing your typing speed.

LESSON 1.5

The Computer Mouse — Use, Buttons, Functions, Mouse Practice

What Is a Mouse?

The mouse is a pointing input device that lets users interact directly with items on the screen — such as icons, buttons, menus, and files — by moving a pointer (cursor) and clicking.

Types of Mice

- **Wired Mouse** — connects to the computer using a USB cable.
- **Wireless Mouse** — connects via Bluetooth or a USB wireless receiver.
- **Trackpad / Touchpad** — a flat touch-sensitive surface built into laptops, used instead of a mouse.
- **Optical / Laser Mouse** — uses a light sensor instead of a rolling ball to track movement.

Parts and Buttons of a Mouse

Part / Button	Function
Left Button	Used for selecting items, clicking buttons, and dragging.
Right Button	Opens a shortcut (context) menu with options related to the item clicked.
Scroll Wheel	Moves up and down a page or document; can often be clicked as a middle button.
Mouse Body	Held in the palm and moved across a surface to move the on-screen cursor.
Sensor (bottom)	Tracks the mouse's movement across the surface it rests on.

Basic Mouse Actions

- **Click (single left-click)** — selects an item or places the cursor.
- **Double-click** — opens a file, folder, or program.
- **Right-click** — opens a context menu with more options (e.g., copy, rename, delete).
- **Click and drag** — holds the left button while moving the mouse to move or select items.
- **Scroll** — rolling the wheel moves the page up or down.
- **Hover** — resting the pointer over an item without clicking, often shows extra information.

Proper Mouse Grip and Posture

- Rest your whole hand gently on the mouse, not just your fingertips.
- Keep your wrist straight, not bent upward or to the side.
- Move the mouse using your whole arm/shoulder for large movements, not just your fingers.
- Keep the mouse at the same height and next to your keyboard to avoid strain.

Mouse Practice Exercises

1. **Pointing Practice:** Move the cursor to each corner of the screen, one at a time, without clicking.
2. **Clicking Practice:** Open the desktop and single-click 5 different icons, one at a time, to select them (without opening).
3. **Double-Click Practice:** Double-click a folder icon to open it, then close it, and repeat 5 times.
4. **Drag-and-Drop Practice:** Click and drag a desktop icon to a new position on the screen.
5. **Right-Click Practice:** Right-click on an empty part of the desktop and identify at least 3 options in the menu that appears.
6. **Scrolling Practice:** Open a long document or webpage and practice scrolling up and down smoothly using the scroll wheel.

Practice Goal

Comfortable mouse control means being able to click precisely on small targets and drag items smoothly without needing to look away from the screen.

LESSON 1.6

Navigating the Computer — Desktop, Icons, Folders

What Is the Desktop?

The **desktop** is the main screen you see after a computer starts up and you log in. It acts as the home base for accessing programs, files, and system settings. The desktop typically includes a background image (wallpaper), icons, and a taskbar or dock.

Key Desktop Elements

Element	Description
Icons	Small pictures representing files, folders, or programs. Double-clicking (or a single click, or
Taskbar / Dock	A bar (usually at the bottom or side of the screen) showing open programs and quick-acces
Start Menu / Applications Menu	A menu that lists installed programs and system settings, opened by clicking the Start butto
System Tray / Menu Bar	Displays the clock, volume, network status, and notification icons.
Wallpaper	The background image or color displayed on the desktop.
Cursor / Pointer	The on-screen arrow or symbol that shows where the mouse is pointing.

Understanding Icons

Icons are visual shortcuts that represent something on the computer. Common icon types include:

- **Program icons** — open an installed application (e.g., a web browser or word processor).
- **Folder icons** — open a folder to reveal the files and folders stored inside it.
- **File icons** — open a specific document, image, video, or other file.
- **Shortcut icons** — a link that points to a program or file located elsewhere; usually marked with a small arrow.

Understanding Folders and Files

A **folder** (sometimes called a directory) is a container used to organize and store files. Folders can also contain other folders, called subfolders, allowing users to build an organized structure similar to a filing cabinet. A **file** is a single piece of saved data, such as a document, photo, song, or video.

Basic Navigation Skills

1. **Opening an item:** Double-click an icon on the desktop to open a program, folder, or file.
2. **Opening the file manager:** Use "File Explorer" (Windows) or "Finder" (macOS) to browse all files and folders on the computer.

3. **Creating a new folder:** Right-click on the desktop or inside a folder, choose "New," then "Folder," and type a name.
4. **Renaming a file or folder:** Right-click the item and choose "Rename," or select it and press the F2 key (Windows).
5. **Moving files:** Click and drag a file from one folder to another, or use Cut (Ctrl + X) and Paste (Ctrl + V).
6. **Deleting an item:** Select the item and press the Delete key, or right-click and choose "Delete"; deleted items are usually sent to the Recycle Bin or Trash first.

Navigation Practice Exercises

1. Create a new folder on the desktop and name it "My Practice Folder."
2. Open the folder, then create two subfolders inside it named "Documents" and "Pictures."
3. Rename "My Practice Folder" to "CS Class Folder."
4. Move one subfolder into the other using drag-and-drop.
5. Open the file manager (File Explorer or Finder) and locate your practice folder using the navigation panel.
6. Delete one of the subfolders and confirm it appears in the Recycle Bin/Trash.

> **Tip**
>
> Keeping a desktop and folder structure organized — with clearly named folders — makes it much faster to find files later and is one of the most useful lifelong computer habits you can build.

Assessment

Introduction to Computer Science — Unit Review

Instructions: Answer all questions to the best of your ability. This assessment covers Lessons 1.0–1.6. Multiple choice questions have only one correct answer unless stated otherwise. The Answer Key appears at the end of this document.

Section A: Multiple Choice

1. What is the correct order of the Information Processing Cycle?

A. Output, Input, Process, Storage
B. Input, Process, Output, Storage
C. Process, Storage, Input, Output
D. Storage, Input, Output, Process

2. Which of the following is an example of an embedded computer?

A. A laptop used for schoolwork
B. A server hosting a website
C. A microwave's built-in control system
D. A desktop gaming computer

3. Which hardware component is considered the "brain" of the computer?

A. RAM
B. Motherboard
C. CPU
D. Power Supply Unit

4. Which of these is an OUTPUT device?

A. Keyboard
B. Microphone
C. Scanner
D. Printer

5. Which type of software manages the computer's hardware and must run before applications can be used?

A. Application software
B. System software
C. Utility games
D. Web browser software

6. Which of the following is an example of application software?

A. Windows operating system

B. A printer driver
C. Microsoft Word
D. Antivirus utility

7. On a keyboard, which key group includes F1 through F12?

A. Alphanumeric keys
B. Function keys
C. Navigation keys
D. Numeric keypad

8. What does the keyboard shortcut Ctrl + Z do?

A. Copies selected text
B. Saves the document
C. Undoes the last action
D. Selects all text

9. Which mouse action is used to open a folder or program?

A. Single left-click
B. Right-click
C. Double-click
D. Scroll

10. What is the main screen shown after a computer starts and a user logs in?

A. The taskbar
B. The desktop
C. The file manager
D. The system tray

Section B: True or False

11. A supercomputer is typically smaller and less powerful than a smartphone.

Answer: ___________

12. RAM is a type of temporary memory that is cleared when the computer is turned off.

Answer: ___________

13. A device driver is an example of system software.

Answer: ___________

14. Right-clicking always opens the selected program immediately.

Answer: ___________

15. A folder can contain other folders, known as subfolders.

Answer: ___________

Section C: Short Answer

16. Name and briefly describe two modern types of computers not including desktops or laptops.

17. List three input devices and three output devices.

18. Explain the difference between system software and application software, and give one example of each.

19. Describe proper hand posture for typing on a keyboard.

20. Explain the steps you would take to create a new folder on the desktop and rename it.

Answer Key

Section A: Multiple Choice

1. B 2. C 3. C 4. D 5. B 6. C 7. B 8. C 9. C 10. B

Section B: True or False

11. False — Supercomputers are among the most powerful computers in the world, used for large-scale scientific computation.
12. True — RAM is volatile memory; its contents are lost when power is removed.
13. True — Device drivers are a type of system software that allow the OS to communicate with hardware.
14. False — Right-clicking opens a context (shortcut) menu; it does not open the program directly.
15. True — Folders can contain subfolders, allowing for organized, nested file structures.

Section C: Short Answer (Sample Guidance)

16. Accept any two of: tablet, smartphone, server, supercomputer, embedded computer — with a correct, brief description of each.
17. Input examples: keyboard, mouse, microphone, scanner, webcam. Output examples: monitor, printer, speakers, headphones.
18. System software manages hardware and runs the computer itself (e.g., Windows OS); application software helps users perform specific tasks (e.g., Microsoft Word).
19. Sit up straight, keep wrists straight and slightly raised, fingers resting on the home row (A-S-D-F / J-K-L-;), eyes on the screen rather than the keyboard.
20. Right-click on the desktop, select "New," then "Folder," type a name and press Enter; to rename, right-click the folder and select "Rename" (or press F2) and type the new name.

Congratulations!

Completing this assessment means you have built a solid foundation in computer science basics — hardware, software, input devices, and navigation. These skills will support you throughout the rest of this course.